

Highly Efficient and Ultra-Low Loss AR-HCF Fusion Splicing with Rapid Rotational Alignment

Wei He⁽¹⁾, Cong Zhang^(1,*), Lipeng Feng⁽²⁾, Di Lin⁽¹⁾, Peng Li⁽³⁾, Lei Zhang⁽³⁾, Jie Luo⁽³⁾, Yuwen Qin⁽¹⁾, and Songnian Fu⁽¹⁾

⁽¹⁾ School of Information Engineering, Guangdong Provincial Key Laboratory of Information Photonics Technology, and Key Laboratory of Photonic Technology for Integrated Sensing and Communication, Ministry of Education of China, Guangdong University of Technology, Guangzhou, 510006, China, zhangcong@gdut.edu.cn

⁽²⁾ China Telecom Research Institute, State Key Laboratory of Optical Fiber and Cable Manufacture Technology, Beijing, 102209, China.

⁽³⁾ State Key Laboratory of Optical Fiber and Cable Manufacture Technology, Yangtze Optical Fiber and Cable Joint Stock Company Ltd., Wuhan 430074, China.

Abstract We propose a discrete-sampling gradient detection algorithm for rapid rotational alignment in anti-resonant hollow-core fiber (AR-HCF) fusion splicing, in order to resolve directional ambiguity, compress the angular search space, and achieve an 0.042 dB average splicing loss with a splicing time less than 55 seconds. ©2026 The Author(s)

Introduction

Anti-resonant hollow-core fibers (AR-HCFs) have emerged as a compelling transmission medium for next-generation, high-capacity optical networks [1-2]. By guiding light in the air, it inherently offers low latency, reduced nonlinearities, and ultra-low transmission loss, with recent breakthroughs demonstrating attenuation as low as 0.05 dB/km in the telecommunications band [3-5].

Benefiting from those inherent advantages, AR-HCFs have a rapid development pace and are being increasingly deployed at in-service networks. However, the unique node-less micro-structured cladding of AR-HCFs introduces a challenge of fusion splicing efficiency. The non-ideal circular profile of fundamental mode makes the coupling efficiency highly sensitive to angular misalignment and require a precise rotational alignment [6]. Meanwhile, the hollow-core characteristic rapidly ingests environmental vapor and gas after cleaving, creating a critically narrow time window for reliable on-field splicing.

To address those challenges of achieving efficient and low-loss splicing, a side-view-based alignment technique was previously extended to AR-HCFs, achieving an average splicing loss of 0.13 dB and a splicing time of 120 seconds within 20 on-field trials [7,8]. By exploiting the N -fold symmetry of AR-HCF, the global angular search space was successfully reduced from 360° down to a $360^\circ/N$ sector. For 5-tube AR-HCF, the angular search space can be reduced from 360° to $360^\circ/5$, leading to a splicing time of 97 seconds with 0.05 dB splicing loss [9].

In the current submission, we propose a discrete-sampling detection algorithm to further compress the angular search space and improve the splicing efficiency. By capturing the gradient trend of the AR-HCF side-view image, the

proposed algorithm can determine the rotational direction and further compress the angular search space from $360^\circ/N$ to $360^\circ/2N$. Therefore, the splicing time are reduced from 97 seconds to 55 seconds for 5-tube AR-HCF and from 83 seconds to 50 seconds for 6-tube AR-HCFs, respectively. Meanwhile, the average splicing loss remains below 0.05 dB for both two kinds of AR-HCFs.

Alignment theory and algorithm

Although our previously proposed symmetry-constrained rotational algorithm (SCRA) [9] successfully restricts the search space of an N -fold symmetric AR-HCF to a $360^\circ/N$ sector, it remains fundamentally limited by directional ambiguity. Without the ability to pre-determine whether a clockwise or counter-clockwise rotation is optimal, the system is forced to scan the entire sector, thereby incurring redundant mechanical motion and excessive image-processing overhead.

Figure 1(a) shows two overlapped 5-tube AR-HCFs, the grey-dashed AR-HCF serves as the reference fiber (Fiber A) and the black-solid AR-HCF (Fiber B) is rotated to achieve the optimal position. The end-view images are included only for clearer illustration, only side-view images are used in the actual alignment process. As shown in Fig. 1(a), when two AR-HCFs are randomly loaded in the fusion splicer, the maximum relative angle difference is 36° . Therefore, the required search range can be reduced from 72° to 36° , if the relative rotational position can be identified.

To solve this directional ambiguity, a discrete-sampling gradient detection method is proposed. Figure 1 (a) and (c) illustrate two distinct scenarios, where the fiber B is oriented clockwise and counterclockwise in relevant to Fiber A, respectively. The similarity evolution between the fiber

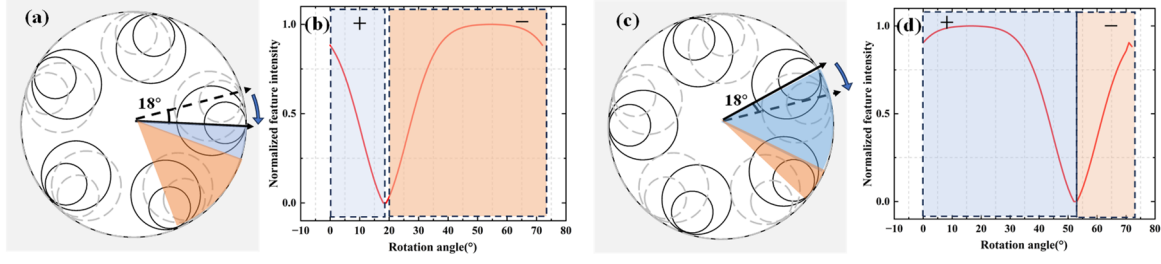


Fig. 1: Schematics of discrete-sampling gradient detection for rotational alignment. Fiber to be rotated (fiber B, black-solid line) is in the (a) Scenarios I: clockwise direction or (c) Scenarios II: counterclockwise direction from reference fiber (fiber A, grey-dashed line). Calculated similarity as a function of Fiber B rotation angle for scenario (b) I and (d) II.

A and rotated fiber B exhibits distinctly different behaviours under both two configurations. As depicted in Fig. 1(b), when the fiber B is positioned 18° clockwise from fiber A, the similarity decreases monotonically with the rotation angle of fiber B, reaching a minimum at a relative angular difference of 36° , and then increases gradually until reaching its maximum when two fibers are perfectly aligned. Alternatively, as shown in Fig. 1(d), when fiber B is offset 18° counterclockwise from fiber A, the similarity initially rises or remains nearly unchanged with increasing rotation angle until two fibers are perfectly aligned. Beyond this point, the similarity declines and reaches a minimum at an angular separation of 36° . Consequently, through discrete sampling during the fiber rotation, the relative rotational orientation can be clearly distinguished.

The detailed operation workflow of the proposed method is summarized in Fig. 2(a). Two kinds of AR-HCFs with different tube units are evaluated. In the coarse alignment stage, the side-view intensity profile of a selected feature region is sampled at several discrete rotational

angles. A local linear fit is then applied to these sampled intensities to estimate the gradient trend with respect to the rotation angle. The sign of this gradient directly indicates whether the rotating fiber is approaching or departing from the target aligned state, thus identifying the correct rotation direction and compressing the angular search space from $360^\circ/N$ to $360^\circ/2N$.

The physical foundation of this direction determination is further illustrated in Fig. 2(b). $I(\theta)$ denote the normalized side-view feature intensity of the rotated fiber as a function of the rotation angle θ . The local gradient can be expressed as

$$G = \frac{dI(\theta)}{d\theta} \quad (1)$$

By evaluating the sign of G , the algorithm judges whether the evolution of sampled intensity is converging toward the reference-aligned state or diverging from it. In this manner, the shortest rotational path can be determined, without scanning the full symmetry sector.

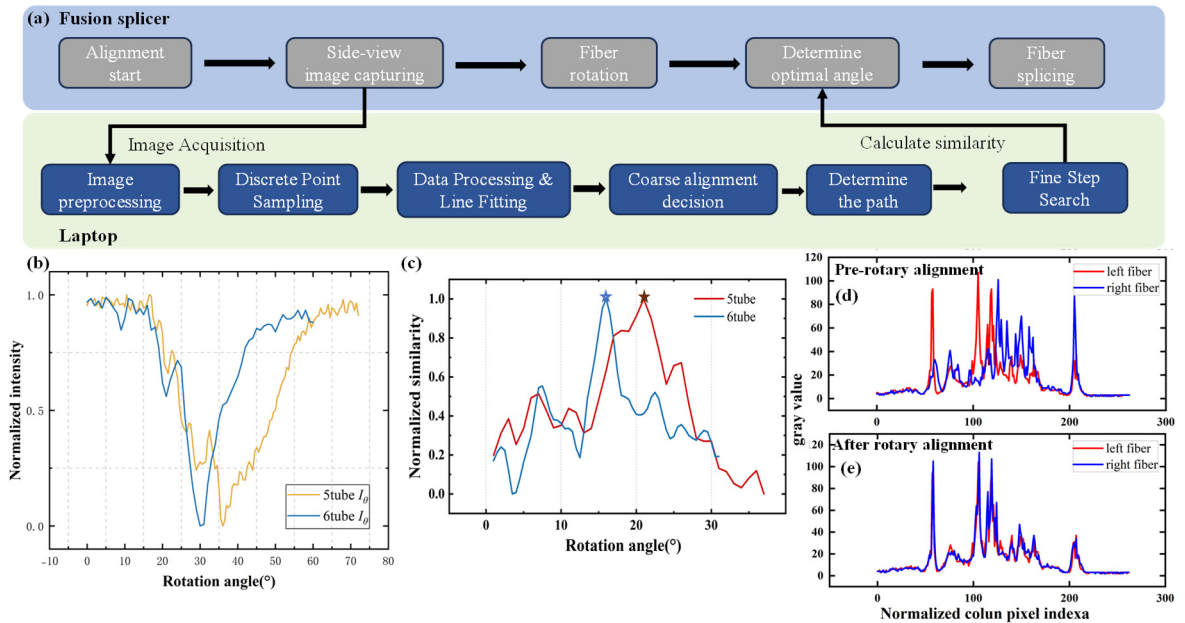


Fig. 2: Discrete sampling gradient detection assisted AR-HCF splicing algorithm. (a) Workflow of the splicing algorithm. (b) Measured side-view feature intensities for 5-tube and 6-tube AR-HCFs during the coarse sampling stage, showing the gradient trend used for rotation-direction determination. (c) High-precision similarity calculation within the compressed search sector. Grayscale intensity profiles of the AR-HCF (d) before and (e) after the algorithm deployment.

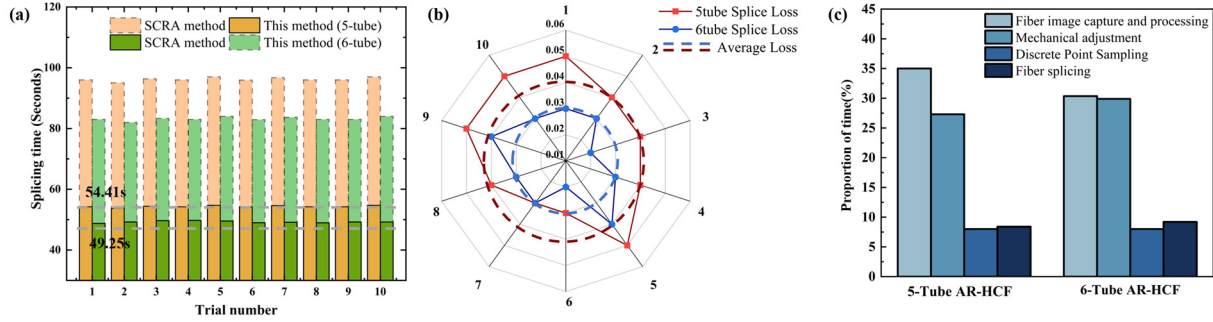


Fig. 3: (a) Splicing time, (b) splicing loss, and (c) time consumption breakdown for 10 independent AR-HCFs splicing trials.

Once the search sector is compressed, the algorithm proceeds to the fine alignment stage within the reduced angular search window. A high-resolution similarity evaluation is subsequently performed to precisely locate the optimal rotational angle. As shown in Fig. 2(c), the normalized similarity curve computed within the compressed search range exhibits a prominent peak, enabling unambiguous identification of the optimal alignment angle. Figure 2(d) depicts the grayscale profile prior to rotational alignment, which corresponds to a random, suboptimal rotational orientation. After applying the algorithm to rotate the AR-HCF to its optimum angular position, the grayscale profile presents a highly periodic and symmetric pattern with well-defined, overlapping peaks.

Fusion splicing characterization and results

To comprehensively evaluate the fusion splicing performance and generalizability of the proposed alignment method, a splicing platform is developed to conduct 10 independent and consecutive splicing trials. A broadband ASE source is utilized alongside SMF-GRIN assemblies at both AR-HCF ends to minimize mode field mismatch [10]. The AR-HCF is cleaved at its midpoint and mounted into a fusion splicer, while a power meter/optical spectrum analyzer (PM/OSA) continuously monitors the optical power. A personal computer (PC) interfaces with the splicer to execute the proposed algorithm, enabling real-time image processing and fully automated rotational alignment. Splicing loss and end-to-end processing times are accurately captured via power variations and timestamps.

As shown in Fig. 3, as for the 5-nested-tube AR-HCF, which operates within the newly compressed 36° search sector, the proposed algorithm achieves a 100% success ratio across all 10 independent trials. The statistical results indicate an average splicing loss of 0.042 dB, with a maximum loss below 0.05 dB. Furthermore, the average end-to-end processing time, including image acquisition, coarse region determination, high-precision fine search, and arc discharge, is reduced from 97 seconds to 54.4 seconds,

bringing a splicing efficiency improvement of 43.3%. This rapid processing speed demonstrates that computational overhead and mechanical scanning latency are effectively minimized by the discrete point sampling mechanism.

To verify the algorithm's adaptability across different cladding geometries, identical batch experiments are performed on 6-tube AR-HCFs. The higher structural symmetry of the 6-tube geometry restricts the initial symmetry period to 60° , which the coarse sampling stage further halves to a 30° search sector. Consequently, the required iterations during the fine step search are proportionally reduced, and the average total splicing time is decreased from 83 seconds to 49.25 seconds, representing a splicing efficiency improvement of 39.7%. Alongside this significant efficiency gain, the tightly constrained 30° search window inherently limited the accumulation of mechanical runout errors, resulting in an exceptionally stable alignment with an average splice loss below 0.03 dB. Figure 3(c) shows the time distribution of each part in a single AR-HCF splicing process, indicating that further splicing efficiency improvement should still focus on the fiber image processing and mechanical adjustment.

Conclusions

We demonstrate a rapid, automated rotational alignment method for the high-precision fusion splicing of AR-HCFs, by utilizing discrete-sampling gradient detection. Once identifying monotonic intensity transitions within the cladding, our proposed approach eliminates directional ambiguity and halves the angular search space. Combined with a fine alignment performed within the compressed search sector, our splicing scheme can achieve a 100% success ratio across all trials. Notably, it reduces average end-to-end splicing times to 54.4 s and 49.25 s for 5- and 6-tube AR-HCFs, respectively, while keeping an ultra-low average loss of less than 0.05 dB. Those results indicate that, the proposed AR-HCF fusion splicing method provides an efficient and reliable solution for the industrial-scale deployment of next-generation AR-HCF communication systems.

Acknowledgements

This work is partially supported by the National Key Research and Development Program of China under Grant (2022YFB2903102), Guangdong Basic and Applied Basic Research Foundation (2024A1515012302), National Natural Science of China (62305071), and Guangdong Introducing Innovative and Entrepreneurial Teams of “The Pearl River Talent Recruitment Program” (2021ZT09X044).

References

- [1] F. Poletti, "Nested antiresonant nodeless hollow core fiber," *Optics Express*, vol. 22, no. 20, pp. 23807–23828, 2014. DOI: [10.1364/OE.22.023807](https://doi.org/10.1364/OE.22.023807)
- [2] R. F. Cregan, B. J. Mangan, J. C. Knight, T. A. Birks, P. S. J. Russell, P. J. Roberts, and D. C. Allan, "Single-mode photonic band gap guidance of light in air," *Science*, vol. 285, no. 5433, pp. 1537–1539, 1999. DOI: [10.1126/science.285.5433.1537](https://doi.org/10.1126/science.285.5433.1537)
- [3] A. I. Adamu, M. R. A. Hassan, Y. Chen, E. N. Fokoua, M. Alonso, H. Sakr, D. J. Richardson, F. Poletti, and M. N. Petrovich, "10.9 km hollow core double nested antiresonant nodeless fiber (DNANF) with 0.33 dB/km loss at 850 nm," in *Proceedings of Optical Fiber Communication Conference (OFC)*, San Diego, CA, USA, 2024, paper M3J.1. DOI: [10.1364/OFC.2024.M3J.1](https://doi.org/10.1364/OFC.2024.M3J.1)
- [4] Y. Ding, P. Li, G. Chen, J. Chu, A. Jia, L. Zhang, J. Luo, "Support tube hollow-core fiber with 0.05 dB/km attenuation," in *Proceedings of European Conference on Optical Communications (ECOC)*, Copenhagen, Denmark, 2025, pp. 1-3. DOI: [10.1109/ECOC66593.2025.11263260](https://doi.org/10.1109/ECOC66593.2025.11263260)
- [5] M. N. Petrovich, E. Numkam Fokoua, Y. Chen, et al., "Broadband optical fibre with an attenuation lower than 0.1 decibel per kilometre," *Nature Photonics*, vol. 19, pp. 1203–1208, 2025. DOI: [10.1038/s41566-025-01391-7](https://doi.org/10.1038/s41566-025-01391-7)
- [6] J. Zeng, C. Zhang, Y. Wang, P. Li, L. Zhang, J. Luo, L. Feng, D. Lin, S. Fu, and Y. Qin, "Low-loss and robust arc-discharge fusion-splicing between anti-resonant hollow-core fibers," *IEEE Photonics Technology Letters*, vol. 37, no. 8, pp. 465–468, 2025. DOI: [10.1109/LPT.2025.3526184](https://doi.org/10.1109/LPT.2025.3526184)
- [7] T. Kremp, Y. Liang, and A. H. McCurdy, "General and reliable azimuthal alignment algorithm for low loss multi-core fiber fusion splicing," *Journal of Lightwave Technology*, vol. 41, no. 12, pp. 3874–3882, 2023. DOI: [10.1109/JLT.2023.3242253](https://doi.org/10.1109/JLT.2023.3242253)
- [8] T. Kremp, Y. Liang, A. H. McCurdy, S. Yoshinaga, and B. J. Mangan, "Fast, reliable and portable low-loss antiresonant hollow-core fiber fusion splicing," in *Proceedings of Optical Fiber Communication Conference (OFC)*, San Diego, CA, USA, 2024, paper M3J.3. DOI: [10.1364/OFC.2024.M3J.3](https://doi.org/10.1364/OFC.2024.M3J.3)
- [9] L. Feng, W. He, C. Zhang, X. Wang, W. Chang, Z. Liu, J. Li, S. Fu, Y. Qin, and C. Zhang, "Fast, Low-Loss, and Field-Deployable Splicing of Anti-Resonant Hollow-Core Fibers," in *Proceedings of Optical Fiber Communication Conference (OFC)*, 2026, paper M1J.4.
- [10] B. Shi, C. Zhang, T. Kelly, X. Wei, M. Ding, M. Huang, S. Fu, F. Poletti, and R. Slavík, "Splicing hollow-core fiber with standard glass-core fiber with ultralow back-reflection and low coupling loss," *ACS Photonics*, vol. 11, no. 8, pp. 3288–3295, 2024. DOI: [10.1021/acsphotonics.4c00424](https://doi.org/10.1021/acsphotonics.4c00424)